Charting the Cognitive Map with Denoising Score Matching

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1 Introduction

The brain represents the external world through internal neural representations. For example, while driving, our brain represents the surrounding spatial environment and the positions of nearby cars. Forming internal representations is a fundamental ability of the brain and serves as the basis for other cognitive functions.

This internal representation is recently referred to as a *cognitive map*. Importantly, a cognitive map is usually different from the corresponding physical map. In spatial navigation, for instance, the physical environment may be a square with a uniform spatial layout. However, the corresponding cognitive map need not be uniform: animals often form biased representations of locations associated with their own niches. The idea of a cognitive map is not limited to spatial locations. Another example is the neural representation of grating orientation. In the physical world, grating images with orientations θ and $\theta + \Delta\theta$ differ only by a rotation. If one measures the Euclidean distance between such images, that distance is independent of θ, reflecting the rotational symmetry of the stimulus space. In this sense, no orientation is intrinsically more special than another in the physical world. In contrast, animals often show better perceptual discrimination for vertical and horizontal orientations than for oblique ones. This suggests that the cognitive map of grating orientation differs from the physical map.

Fisher information is a fundamental quantity for characterizing such cognitive maps. At a high level, the Fisher information at a stimulus value θ quantifies the upper bound on how accurately θ can be decoded from neural activity. Therefore, Fisher information tells us how well the brain represents a particular stimulus value.

From another perspective, Fisher information measures the local distance between nearby stimulus values in neural representation space. Returning to the grating example, although the Euclidean distance between physical images with nearby orientations is independent of orientation, the neural distance—as quantified by Fisher information—can be larger near vertical and horizontal orientations than near oblique ones. This geometric perspective directly connects Fisher information to the notion of a cognitive map: Fisher information defines a local metric on the stimulus manifold represented by the neural population. An important advantage of Fisher information over the Euclidean distance is that Fisher information is invariant to the subjective choice of coordinate system. Fisher information therefore reflects intrinsic properties of the neural representation.

However, estimating Fisher information from neural population activity is hard, arguably because neural responses are both high-dimensional and noisy. Existing methods for estimating Fisher information can be broadly divided into two categories.

First, there are model-based methods, which assume a specific statistical model for neural activity, such as Poisson spiking or an exponential-family model. In practice, however, neural population activity is shaped by complex inter-neuron interactions, and the true statistical model is typically unknown.

Second, there are model-free methods. A central idea in this category is to train a classifier to distinguish between two nearby stimulus values based on neural activity x, and then use the classifier performance to estimate Fisher information. This approach is natural when stimulus values are discretized and repeated trials are available for each stimulus. However, many neuroscience experiments (such as navigation tasks)—especially increasingly common naturalistic experiments—do not provide repeated trials at discrete stimulus values; for example, an animal's spatial location varies continuously during navigation. To apply classical model-free methods in such settings, one must first discretize the continuous stimulus into bins, which introduces subjective choices such as bin width. Moreover, such discretization fails to exploit the continuity of the stimulus space, where nearby observations may contain useful information for estimating Fisher information.

More recently, score-matching-based methods have been proposed for Fisher information estimation. Score matching is a generative framework closely related to diffusion models, which have achieved remarkable success in generating high-dimensional data such as images, audio, and video. Recent work has adapted score-based techniques to estimate Fisher information. However, these methods often require modifying the score matching method itself, such as latent-variable inference or simulator-specific objectives. As a result, they cannot always directly leverage the broad and rapidly developing family of existing score-matching methods. Furthermore, their applicability to high-dimensional, noisy neural data—especially discrete spiking data—remains unclear.

In this manuscript, we propose a simple alternative. Using Bayes' rule, we express the Fisher score in terms of Stein score which can be directly estimated by score-matching methods without any modification. In this way, our approach can immediately benefit from the broad and rapidly advancing score-matching literature. Moreover, the proposed method can be applied directly to spiking data, without converting spikes into continuous firing rates or imposing a Poisson assumption.

2 Preliminary results

2.1 Bayes' rule to express Fisher score in terms of Stein score

Score matching is a method to estimate the Stein score, however, is different from Fisher score. Therefore we express Fisher score in terms of Stein score using Bayes' rule.

$$\nabla_\theta \log p(x \mid \theta) = \nabla_\theta \log p(\theta \mid x) - \nabla_\theta \log p(\theta), \tag{1}$$

where $\theta \in \mathbb{R}^d$ is the stimulus parameter, such as grating orientation or spatial position; $x \in \mathbb{R}^n$ is the neural population activity; $\nabla_\theta \log p(x \mid \theta)$ is the Fisher score, $\nabla_\theta \log p(\theta \mid x)$ is the Stein score, and $\nabla_\theta \log p(\theta)$ is the score of the stimulus prior. In controlled experiments the prior may be known (e.g. uniform orientations); otherwise it can be learned from samples of θ alone. When $d \ll n$, the posterior score $\nabla_\theta \log p(\theta \mid x)$ can be estimated with conditional score matching; the prior score $\nabla_\theta \log p(\theta)$ can be estimated with a separate unconditional score model on θ.

2.2 Conditional denoising score matching

Here we focus on describing conditional denoising score matching for estimating $\nabla_\theta \log p(\theta \mid x)$. A typical way of using conditional denoising score matching is to let it learn and generate high-dimensional data; however, here we use it in an opposite way, the condition x is high-dimensional and the target θ is typically low-dimensional. Though, this does not change the mathematical nature of the problem – any conditional score matching method can be used to estimate $\nabla_\theta \log p(\theta \mid x)$. It is a plug-in component of our Fisher information estimation framework.

For better understanding, here we describe a concrete implementation of conditional denoising score matching. The spirit of denoising score matching is denoising. One adds Gaussian noise to a variable of interest and fits a network so that its output matches the noise that was injected; equivalently, the trained network approximates the Stein score—the gradient of the log-density—of a slightly smoothed distribution.

Let θ denote the stimulus parameter and let x denote an observation drawn from the conditional model $p(x \mid \theta)$. We sample a log-noise level $\log \sigma$ uniformly on an interval $[\log \sigma_{\min}, \log \sigma_{\max}]$, draw a standard normal scalar $z \sim \mathcal{N}(0, 1)$, and form the noisy latent $\tilde{\theta} = \theta + \sigma z$. We parameterize the score by a neural network s_ϕ with trainable weights ϕ and train it by minimizing the noise-conditional denoising objective

$$\mathcal{L}(\phi) = \mathbb{E}\left[\left\|\sigma\, s_\phi(\tilde{\theta}, x, \sigma) + z\right\|^2\right], \tag{2}$$

where the expectation is taken over (θ, x) drawn from the generative process, over the random choice of σ, and over z. Here $\mathcal{L}(\phi)$ is the scalar loss to be minimized, ϕ collects all network parameters, and $s_\phi(\tilde{\theta}, x, \sigma)$ is the network output intended to approximate the score of a σ-smoothed conditional posterior, in the θ-direction, at $(\tilde{\theta}, x, \sigma)$: namely $\nabla_{\tilde{\theta}} \log p_\sigma(\tilde{\theta} \mid x)$, where $p_\sigma(\cdot \mid x)$ is the density of $\tilde{\theta} = \theta + \sigma z$. As $\sigma \to 0$, $p_\sigma(\cdot \mid x)$ converges to $p(\cdot \mid x)$ and $\nabla_{\tilde{\theta}} \log p_\sigma(\tilde{\theta} \mid x)$ converges to the Stein score $\nabla_\theta \log p(\theta \mid x)$. Together with $\nabla_\theta \log p(\theta)$, estimated by a second denoising score model on θ alone (see Method), we obtain the Fisher score via equation 1.

2.3 Preliminary experiments

We illustrate the full pipeline on a Gaussian toy model with scalar θ and two-dimensional observations $x \in \mathbb{R}^2$: cosine tuning means $\mu(\theta)$ and a θ-dependent covariance (heteroscedastic, correlated noise). We draw $n_{\text{total}} = 4000$ pairs with $\theta \sim \text{Uniform}[-3, 3]$, use a 70/30 train/eval split (2800/1200), and train (i) a conditional score network $s_\phi(\tilde{\theta}, x, \sigma)$ for the posterior score, (ii) an unconditional score network $s_\psi(\tilde{\theta}, \sigma)$ for the prior on θ, and (iii) local decoder baselines per θ bin. Figure 1 compares Fisher information vs. θ for analytic ground truth, posterior-only score at $\sigma_{\min}$, likelihood score $s_\phi - s_\psi$ at $\sigma_{\min}$, and the decoder (shaded bands: reported uncertainty).

Numerical metrics vs. ground truth (35 valid bins) are:

Method vs. GT	RMSE	MAE	Correlation
Posterior score ($\sigma_{\min}$)	8.525	5.489	0.918
Likelihood score ($s_{\text{post}} - s_{\text{prior}}$, $\sigma_{\min}$)	8.039	5.246	0.930
Decoder	21.417	13.738	0.727

3 Future possible extensions

Planned directions include comparisons on additional toy models with known ground truth (including spiking observations), and applications to neural data in visual and navigation tasks to characterize representation bias in cognitive-map terms.

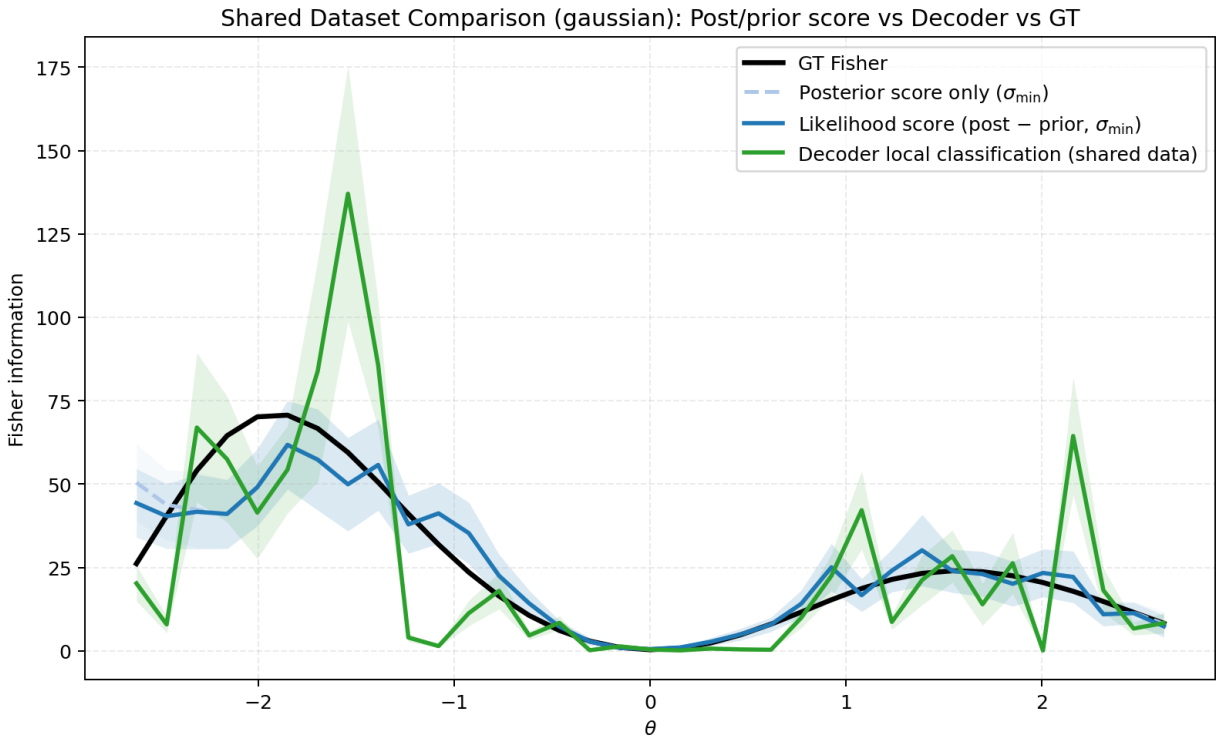

Figure 1: Gaussian toy ($n_{\text{total}} = 4000$, $x \in \mathbb{R}^2$). Fisher information vs. θ: ground truth (black), posterior score at $\sigma_{\min}$ (dashed blue), likelihood score $s_{\text{post}} - s_{\text{prior}}$ at $\sigma_{\min}$ (solid blue), local decoder (green).

4 Method

Data-generating model

The latent parameter is sampled as

$$\theta \sim \text{Uniform}[-3, 3],$$

and observations are generated from a Gaussian conditional family

$$x \mid \theta \sim \mathcal{N}\big(\mu(\theta), \Sigma(\theta)\big), \qquad x \in \mathbb{R}^2.$$

Each coordinate of $\mu(\theta)$ uses a cosine tuning curve with a fixed phase; the covariance $\Sigma(\theta)$ varies smoothly with θ (coordinate-wise standard deviations and correlation), yielding heteroscedastic correlated Gaussian observations. The experiment in Section 2 uses $n_{\text{total}} = 4000$, seed 7, and train fraction 0.7 (2800 train, 1200 eval).

Posterior and prior score models

Posterior (conditional) score. We train $s_\phi(\tilde{\theta}, x, \sigma)$ with continuous noise-conditional denoising score matching. For each pair (θ, x) from the data-generating process,

$$\log \sigma \sim \text{Uniform}(\log \sigma_{\min}, \log \sigma_{\max}), \qquad z \sim \mathcal{N}(0, 1), \qquad \tilde{\theta} = \theta + \sigma z.$$

The objective is

$$\mathcal{L}(\phi) = \mathbb{E}\left[\left(\sigma s_\phi(\tilde{\theta}, x, \sigma) + z\right)^2\right].$$

Noise levels are tied to the empirical spread of θ on the score *fit* split: with $\hat{s}_\theta = \mathrm{std}(\theta_{\mathrm{fit}}) = 1.7273$,

$$\sigma_{\min} = 0.01\,\hat{s}_\theta = 0.01727, \qquad \sigma_{\max} = 0.25\,\hat{s}_\theta = 0.4318,$$

with 12 logarithmically spaced evaluation σ values between these extremes. Training uses a 15% validation holdout from the score-training pool (2380 fit / 420 validation), early stopping with patience 1000, minimum improvement 10^{-4}, exponential moving average (EMA) smoothing of the validation loss with $\alpha = 0.05$, and restoration of the best checkpoint.

Prior (unconditional) score. We train a separate network $s_\psi(\tilde{\theta}, \sigma)$ on θ samples only, with the same noise injection on $\tilde{\theta}$ and the denoising objective

$$\mathcal{L}(\psi) = \mathbb{E}\left[\left(\sigma s_\psi(\tilde{\theta}, \sigma) + z\right)^2\right],$$

and analogous early stopping (same patience, tolerance, EMA $\alpha = 0.05$). This approximates the prior score $\nabla_\theta \log p(\theta)$ in the smoothed sense used by DSM.

Score-based Fisher estimators

Let σ_* denote the smallest evaluation noise level (here 0.01727). For each evaluation pair (θ, x) we form posterior and prior score predictions at σ_* and define the likelihood score $\hat{s}_{\mathrm{lik}} = s_\phi(\theta, x, \sigma_*) - s_\psi(\theta, \sigma_*)$. Fisher is estimated by binning θ into 35 equal-width bins on $[\theta_{\mathrm{low}} + m, \theta_{\mathrm{high}} - m]$ with margin $m = 0.30$, requiring at least 10 points per bin, and averaging squared scores within bins:

$$\widehat{I}_{\mathrm{post}}(\theta_0) = \mathbb{E}\big[s_\phi(\theta, x, \sigma_*)^2 \,\big|\, \theta \in \mathrm{bin}(\theta_0)\big], \qquad \widehat{I}_{\mathrm{lik}}(\theta_0) = \mathbb{E}\big[\hat{s}_{\mathrm{lik}}(\theta, x)^2 \,\big|\, \theta \in \mathrm{bin}(\theta_0)\big].$$

The primary reported score curve is $\widehat{I}_{\mathrm{lik}}$ (posterior minus prior); $\widehat{I}_{\mathrm{post}}$ is kept as an ablation.

Decoder Fisher baseline

For each bin center θ_0, define two nearby parameters

$$\theta_+ = \theta_0 + \frac{\varepsilon}{2}, \qquad \theta_- = \theta_0 - \frac{\varepsilon}{2}, \qquad \varepsilon = 0.12.$$

A local classifier is trained to distinguish samples near θ_+ and θ_- using neighborhood bandwidth 0.10 in θ. If $\ell(x)$ is the classifier logit, then for small ε,

$$\ell(x) \approx \varepsilon\, \partial_\theta \log p(x \mid \theta_0),$$

and the decoder-based Fisher estimator is

$$\widehat{I}_{\mathrm{dec}}(\theta_0) = \frac{1}{\varepsilon^2}\, \mathbb{E}\big[\ell(x)^2\big].$$

Decoder training uses validation fraction 0.15, minimum validation class size 10, patience 100, minimum improvement 10^{-4}, EMA smoothing with $\alpha = 0.2$, up to 80 epochs, and best-epoch restoration.

Ground-truth Fisher and evaluation protocol

For the Gaussian conditional model, we evaluate ground truth analytically,

$$I_{\mathrm{gt}}(\theta) = \mu'(\theta)^\top \Sigma(\theta)^{-1} \mu'(\theta) + \frac{1}{2} \operatorname{tr}\big[\Sigma(\theta)^{-1} \Sigma'(\theta) \Sigma(\theta)^{-1} \Sigma'(\theta)\big] \,,$$

with derivatives induced by the chosen mean and covariance parameterization (equivalently the Fisher-trace definition).

Estimated curves are compared to I_{gt} over the 35 bins using RMSE, MAE, and Pearson correlation, as in Figure 1.